

SOURAVA KUMAR BEHERA

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Souravakb24

PROFESSIONAL SUMMARY

NLP Research Engineer with 1.5 years of experience building AI systems. Passionate in working on LLMs including continual pre-training, LoRA fine-tuning, Agentic frameworks, and NMT, with research publications and real-world deployments in domains like NMT, RAG, Document OCR at scale.

EDUCATION

Indian Institute of Technology, Mandi

M.Tech. Research, School of Computing & Electrical Engineering

Jan 2025 – Present

Institute of Technical Education and Research, Bhubaneswar

B.Tech. in Computer Science and Engineering

Aug 2020 – Jun 2024

TECHNICAL SKILLS

Languages & Tools: Python, Java, Git, GitHub

Core Expertise: NLP, LLMs, Deep Learning, Agentic Systems, RAG, NMT, NL2SQL

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, FastAPI

GenAI Stack: LLMs, LoRA/PEFT, LLaMA, BERT, BART, VLMs, vLLM

Specialisations: LLM Pre-training & Fine-tuning, OCR Systems, Low-resource Indic NLP

PROFESSIONAL EXPERIENCE

BharatGen

Project Associate — NLP Research & Development / Indian Institute of Technology, Mandi, H.P.

Mar 2025 – May 2026

Agentic RAG Platform for Government Knowledge Systems

- Designed an agentic pipeline with semantic chunking, enabling grounded multi-document retrieval across 500–1,000 government policy and technical documents.
- Enabled automated figure reconstruction using Agentic based reasoning to generate SVG/Python code from natural-language descriptions & figure geometry, reducing manual effort by ~60%.
- Built API-based scalable task pipelines supporting high-throughput document ingestion (30+ docs/hr) and real-time retrieval.
- Developing an AI-powered dynamic ontology pipeline for automated knowledge graph construction using multi-agent LLMs.

LLM Domain Adaptation

- Curated an 860M token domain-specific corpus from books, research papers, and educational videos using VLM-based OCR pipelines with vLLM inference.
- Performed continual pre-training on a 2.9B parameter model, outperforming 7B+ general-purpose models on domain-specific benchmarks (+4.2 F-score on held-out QA evaluation).
- Created an SFT dataset with LLM-generated outputs and human supervision; applied LoRA fine-tuning, reducing error rate by 35% on MCQ test set and 16% overall across all tasks.

Reverie Language Technologies (Reliance Venture)

Software Engineer R&D / Bengaluru, Karnataka

Jul 2024 – Jan 2025

Odia NMT | OSWAS Government Integration

- Developed an Odia NMT system achieving BLEU 34.8 vs. Google Translate's 31.2, deployed as a production API for real-time government document translation.
- Implemented a data mixing strategy to handle selective transliteration on translation tasks.

Automated Document Translation Pipeline

- Built a document-structure-aware automated pipeline using Document Parsers and NMT models to translate English documents into Indic languages.
- Preserved alignment in document post-translation, eliminating manual reformatting and reducing human annotation effort by 70%.

Indic NLP Systems & Banking-Domain RAG

- Built Indic NER for 11 Indian languages using BERT, expanding entity classes to music, books, and movies domains
- Developed a Banking-domain RAG chatbot with hybrid retrieval and query re-ranking.

Reverie Language Technologies (Reliance Venture)

Intern R&D / Remote

Sep 2023 – Jun 2024

- Fine-tuning of SLMs (1B - 3B) and Seq2Seq models for Hindi grammar checker & paraphraser.
- Scraped and cleaned 1,000+ movie subtitle files across multiple Indian languages for parallel corpus datasets using sentence alignment techniques.

PUBLICATIONS

- HiLearners: Non-Native Spoken Hindi Error Correction** — IJCNLP-AACL 2025
- Multi-Feature Graph Convolution Network for Hindi OCR Verification** — BHASHA Workshop, IJCNLP-AACL 2025 [Best Paper Award]

CERTIFICATIONS & ACHIEVEMENTS

- Technical Mentor at Career Development Hub (CDH ITER): DSA & Machine Learning
- Completed Geeks for Geeks 11-week DSA Workshop; solved 450+ DSA problems
- Participant in Hack For Climate Hackathon (Indian Oil) | Winner — Regional Level Science Exhibition